

ECE 332 — Homework 4: Generated Answers from HW4 Cheat Sheet

Each solution shows the cheat sheet snippet used, then the answer.

Symbol reference (used throughout):

- l — antenna physical length (m)
- $\lambda = c/f$ — wavelength (m); $c = 3 \times 10^8$ m/s
- I_0 — peak current at antenna feed (A)
- R — distance from antenna to observation point (m)
- θ — angle from antenna axis (rad or deg)
- $\eta_0 = 120\pi \approx 377 \Omega$ — intrinsic impedance of free space
- $k = 2\pi/\lambda$ — wavenumber (rad/m)
- $S(R, \theta)$ — average power density radiated (W/m²)
- $F(\theta) = S(\theta)/S_{\max}$ — normalized radiation intensity (unitless)
- β — half-power beamwidth (rad or deg)
- Ω_p — pattern solid angle (sr)
- $D = 4\pi/\Omega_p$ — directivity (unitless, often dB)
- $A_e = \lambda^2 D / (4\pi)$ — effective aperture area (m²)
- A_p — physical cross-section area (m²)
- G_t, G_r — antenna gains (linear)
- P_t, P_r — transmitted / received power (W)

Key shortcuts: *Hertzian dipole* ($l/\lambda < 1/50$) $\Rightarrow D = 1.5$. *Half-wave dipole* ($l = \lambda/2$) $\Rightarrow D = 1.64$, $R_{\text{rad}} = 73 \Omega$. *3 dB* $\Rightarrow$ *linear gain of 2*.

Problem 1 — Hertzian Dipole, Power Density at 5 km

8 pts

Setup: A 1-m long dipole excited by a 1 MHz current with amplitude $I_0 = 12$ A. Find the average power density radiated at $R = 5$ km, $\theta = 45^\circ$.

DIPOLE TYPE BY LENGTH

Compute first $\lambda = c/f$, then ratio l/λ

l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D = 1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave short-cut, $D = 1.64$, $R_{\text{rad}} = 73 \Omega$
$= 1$	Full-wave	ARB, $D = 2.41$, $R_{\text{rad}} = 199 \Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — wrong shape.

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors small $l \ll \lambda$
 $\tilde{E}_\theta = \frac{j I_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta, \quad \tilde{H}_\phi = \tilde{E}_\theta / \eta_0$
Power density if $l/\lambda < 1/50$; far field
 $S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta = S_{\text{max}} \sin^2 \theta \text{ (W/m}^2\text{)}$
 $S(R, \theta)$ — avg. pwr dens. at (R, θ) (W/m²)
 l — antenna length (m, the rod itself)
 I_0 — peak current in dipole (A)
 R — distance from antenna to obs. point (m)
 θ — angle from dipole axis (rad/°)
 $k = 2\pi/\lambda$ — wavenumber (rad/m)
 $\eta_0 = 120\pi \approx 377 \Omega$ — free-space impedance
Max direction broadside $\theta_{\text{max}} = \pi/2$
Total radiated power integrate over 4π
 $P_{\text{rad}} = \frac{\eta_0 \pi}{3} \left(\frac{I_0 l}{\lambda} \right)^2 \text{ (W)}$
Directivity exact $D_{\text{Hertz}} = 1.5$ (1.76 dB)

COMMON DIPOLE PARAMETERS

Type	D	D_{dB}	R_{rad} (Ω)
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2 (l/\lambda)^2$
Half-wave	1.64	2.15	73.2
$\lambda/4$ monopole	1.64	2.15	36.6
$l = \lambda$	2.41	3.82	199

Half-wave: $P_{\text{rad}} = 36.6 I_0^2 \text{ W}$. $\lambda/4$ monopole (above ground); same patt. as half-wave but $P_{\text{rad}}, R_{\text{rad}}$ halved.

Step 1 — Classify the dipole length. Compute the wavelength to compare with l :

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^6} = 300 \text{ m}$$

$$\frac{l}{\lambda} = \frac{1}{300} < \frac{1}{50} \Rightarrow \text{Hertzian dipole}$$

The DIPOLE TYPE BY LENGTH section says short ($l/\lambda < 1/50$) means we can use the Hertzian $S(R, \theta)$ formula directly.

Step 2 — Apply the Hertzian power density formula. From HERTZIAN DIPOLE — $S(R, \theta)$:

$$S(R, \theta) = \left(\frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \right) \sin^2 \theta$$

Step 3 — Plug in numbers. $\eta_0 = 120\pi$, $k = 2\pi/\lambda = 2\pi/300$ rad/m, $I_0 = 12$ A, $l = 1$ m, $R = 5000$ m, $\theta = \pi/4$:

$$S = \frac{(120\pi)(2\pi/300)^2(12)^2(1)^2}{32\pi^2(5 \times 10^3)^2} \sin^2(\pi/4)$$

$$\text{Numerator: } (120\pi)(4\pi^2/9 \times 10^4)(144) = (120\pi)(144)(4\pi^2)/(9 \times 10^4)$$

$$= 120 \cdot 144 \cdot 4 \cdot \pi^3 / (9 \times 10^4) = 69120\pi^3 / (9 \times 10^4)$$

$$\text{Denominator: } 32\pi^2 \cdot 2.5 \times 10^7. \text{ And } \sin^2(\pi/4) = 1/2.$$

Working it out cleanly:

$$S(5 \text{ km}, 45^\circ) = 1.51 \times 10^{-9} \text{ W/m}^2$$

Sanity: a 12 A current at 1 MHz across a 1-m antenna at 5 km radiates 1.5 nW/m². The $1/R^2$ falloff dominates: most of the radiated power scatters into space, only a tiny fraction reaches any one observation point.

Problem 2 — Beam Parameters from Normalized Pattern

15 pts

Setup: $F(\theta) = \exp(-20\theta^2)$ for $0 \leq \theta \leq \pi$ (θ in radians). Find: (a) half-power beamwidth β , (b) pattern solid angle Ω_p , (c) directivity D .

BEAM PARAMETERS — β , Ω_p , D Step 1 — Normalize *always start here*

$$F(\theta) = S(\theta)/S_{\max} \text{ (unitless)}$$

Step 2 — HPBW β *half-power beamwidth, full angular width at -3 dB*Set $F(\theta) = 0.5$, solve for $\theta \rightarrow \theta_{1/2}$

$$\beta = 2\theta_{1/2} \text{ (symmetric beams, rad or } ^\circ)$$

Gaussian:

$$F(\theta_{1/2}) = e^{-a\theta_{1/2}^2} = 0.5 \Rightarrow \theta_{1/2} = \sqrt{\ln 2/a}$$

Step 3 — Pattern solid angle Ω_p *units: sr*

$$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$$

No ϕ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta \, d\theta$.*TI-36X Pro: [2nd] [d/dx] for $\int_a^b$, set RAD mode, multiply result by 2π .**Narrow-beam Gaussian shortcut ($\sin \theta \approx \theta$, $\rightarrow \infty$):*

$$\Omega_p \approx \frac{\pi}{a} = \frac{\pi \theta_{1/2}^2}{\ln 2} \quad (F = e^{-a\theta^2})$$

Step 4 — Directivity D

$$D = \frac{4\pi}{\Omega_p} \text{ (unitless)} \quad D_{\text{dB}} = 10 \log_{10} D$$

*Alt. with two orthogonal HPBW: $D \approx 4\pi/(\beta_{xz}\beta_{yz})$.*Step 5 — Gain (if asked) $\xi = \text{rad. eff.}$, $0 < \xi \leq 1$

$$G = \xi D \quad G_{\text{dB}} = 10 \log_{10} G$$

Lossless / ideal antenna $\Rightarrow G = D$.

(a) **Half-power beamwidth β (5 pts).** From BEAM PARAMETERS Step 2, the original cheat sheet equations:

$$F(\theta_{1/2}) = 0.5 \Rightarrow \text{solve for } \theta_{1/2}$$

$$\beta = 2\theta_{1/2} \text{ (symmetric beams)}$$

Plug in $F(\theta) = \exp(-20\theta^2)$:

$$\exp(-20\theta_{1/2}^2) = 0.5 \Rightarrow -20\theta_{1/2}^2 = \ln(0.5)$$

$$\theta_{1/2} = \sqrt{-\ln(0.5)/20} = \sqrt{0.693/20} = 0.186 \text{ rad}$$

That's the half-angle from the peak to one -3 dB point. Since the beam is symmetric about $\theta = 0$, the *full half-power beamwidth* is twice that:

$$\boxed{\beta = 2\theta_{1/2} = 0.372 \text{ rad} = 21.31^\circ}$$

(b) **Pattern solid angle Ω_p (5 pts).** From BEAM PARAMETERS Step 3:

$$\Omega_p = \iint_{4\pi} F(\theta) \sin \theta \, d\theta \, d\phi = \int_0^{2\pi} \int_0^\pi \exp(-20\theta^2) \sin \theta \, d\theta \, d\phi$$

F has no ϕ dependence, so $\int_0^{2\pi} d\phi = 2\pi$:

$$\Omega_p = 2\pi \int_0^\pi \exp(-20\theta^2) \sin \theta d\theta$$

This integral has no closed form; evaluate numerically on the TI-36X Pro.

TI-36X Pro keystrokes (definite integral):

1. Press [2nd] $\rightarrow$ [d/dx] ($\rightarrow$ definite-integral template appears, shown as $\int_{\square}^{\square} \square d\square$).
2. **Lower limit:** type 0, then $\rightarrow$ arrow to the upper limit.
3. **Upper limit:** press [π], then arrow into the integrand box.
4. **Integrand:** type $e^{(-20 \times x^2) \times \sin(x)}$ (use [e^x] for the exp, [x^2] for the square, [$\sin$] for the sine).
5. **Variable:** arrow to the $d\square$ slot, type x .
6. Press [enter] $\rightarrow$ calculator returns ≈ 0.024792 .
7. Multiply by 2π : $\times [\pi] \times 2$ [enter] $\rightarrow \approx 0.15578$.

$$\Omega_p \approx 0.1558 \text{ sr}$$

Keep the unrounded calculator value for part (c). Textbook rounds Ω_p to 0.156 first and gets $D = 80.55$; using the full precision gives $D \approx 80.67$. Make sure the calculator is in RADIANS mode ([mode] $\rightarrow$ RAD) since θ in the exponent is bare — degree mode will give nonsense.

(c) Directivity D (5 pts). From BEAM PARAMETERS Step 4:

$$D = \frac{4\pi}{\Omega_p} = \frac{4\pi}{0.15578}$$

$$D = 80.67$$

Using textbook's rounded $\Omega_p = 0.156$: $D = 4\pi/0.156 = 80.55$. Use the unrounded value to avoid this rounding error.

Sanity: $D \approx 80$ is highly directive (much narrower than half-wave dipole's 1.64). The narrow Gaussian-like beam ($\beta \approx 21^\circ$) concentrates radiation into a small solid angle, so directivity is large.

Problem 3 — $\lambda/4$ Dipole Radiation Pattern

15 pts

Setup: Dipole of length $l = \lambda/4$. Find: (a) directions of maximum radiation, (b) expression for $S_{\max}$, (c) plot of normalized pattern $F(\theta)$.

ARBITRARY-LENGTH DIPOLE — $S(\theta)$ Power density at R any l

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right]^2$$

Normalized pattern *dimensionless* $F(\theta) = S(\theta)/S_{\max}$ Quarter-wave $l = \lambda/4$ $\pi l/\lambda = \pi/4$, $\cos(\pi/4) = \sqrt{2}/2$

$$\frac{S(\theta)}{S_0} = \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2, \quad S_0 = \frac{15I_0^2}{\pi R^2}$$

$$S_{\max} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858 S_0$$

Half-wave $l = \lambda/2$ $\pi l/\lambda = \pi/2$, *shortcut form*

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \frac{\cos^2\left(\frac{\pi}{2} \cos \theta\right)}{\sin^2 \theta}$$

 $D = 1.64$, $R_{\text{rad}} = 73 \Omega$, $\theta_{\max} = \pi/2$.

COMMON DIPOLE PARAMETERS

Type	D	D_{dB}	$R_{\text{rad}} (\Omega)$
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2 (l/\lambda)^2$
Half-wave	1.64	2.15	73.2
$\lambda/4$	1.64	2.15	36.6
monopole			
$l = \lambda$	2.41	3.82	199

Half-wave: $P_{\text{rad}} = 36.6 I_0^2 W$. $\lambda/4$ monopole (above ground): same patt. as half-wave but P_{rad} , R_{rad} halved.

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

$^\circ$	rad	$\sin \theta$	$\cos \theta$	$\sin^2 \theta$	$\cos^2 \theta$
0	0	0	1	0	1
30	$\pi/6$	1/2	$\sqrt{3}/2$	1/4	3/4
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1/2	1/2
60	$\pi/3$	$\sqrt{3}/2$	1/2	3/4	1/4
90	$\pi/2$	1	0	1	0
180	π	0	-1	0	1

Convert: $\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \pi/180$. Antenna formulas usually take θ in rad.

General arbitrary-length dipole pattern. From ARBITRARY-LENGTH DIPOLE — $S(\theta)$:

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right]^2$$

Plug in $l = \lambda/4$, so $\pi l/\lambda = \pi/4$:

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \cos(\pi/4)}{\sin \theta} \right]^2 = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$$

Let $S_0 = 15I_0^2/(\pi R^2)$ be the constant prefactor.

(a) Direction of maximum radiation (5 pts). Three-step argument: (i) endpoints via L'Hôpital, (ii) symmetry about $\pi/2$, (iii) sampling to confirm single peak.

Step (i) — endpoints $\theta = 0, \pi$ via L'Hôpital. Inside the squared bracket: $f(\theta) = \cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}$ over $g(\theta) = \sin \theta$.

At $\theta = 0$: $f(0) = \cos(\pi/4) - \sqrt{2}/2 = 0$ and $g(0) = 0 \Rightarrow 0/0$ form. Apply L'Hôpital:

$$f'(\theta) = \frac{\pi}{4} \sin\left(\frac{\pi}{4} \cos \theta\right) \sin \theta, \quad g'(\theta) = \cos \theta$$

$$\lim_{\theta \rightarrow 0} \frac{f'(\theta)}{g'(\theta)} = \frac{(\pi/4) \sin(\pi/4) \cdot 0}{1} = 0$$

So $S(0)/S_0 = 0^2 = 0$. Same algebra at $\theta = \pi$ (with $g'(\pi) = -1$) gives $S(\pi) = 0$.

Step (ii) — symmetry about $\theta = \pi/2$. Substitute $\theta \rightarrow \pi - \theta$: $\cos(\pi - \theta) = -\cos \theta$, but $\cos$ is even so $\cos((\pi/4) \cos(\pi - \theta)) = \cos((\pi/4) \cos \theta)$. And $\sin(\pi - \theta) = \sin \theta$. So $S(\theta) = S(\pi - \theta)$, pattern mirrors about $\pi/2$.

Step (iii) — sample values to confirm single peak (not M-shape). Compute $S(\theta)/S_0$ for a few angles:

θ	30°	60°	90°
S/S_0	0.020	0.063	0.086

Monotone rise to 90° , then symmetric fall. Single peak at the center.

$$\theta_{\max} = \pi/2 = 90^\circ$$

(broadside to the dipole axis, same as half-wave dipole.)

(b) $S_{\max}$ (5 pts). Evaluate $S(\pi/2)$:

$$S_{\max} = S(\pi/2) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi}{4} \cdot 0) - \frac{\sqrt{2}}{2}}{\sin(\pi/2)} \right]^2 = \frac{15I_0^2}{\pi R^2} \left(1 - \frac{\sqrt{2}}{2} \right)^2$$

Numerically $(1 - \sqrt{2}/2)^2 = 0.0858$:

$$S_{\max} = 0.0858 \frac{15I_0^2}{\pi R^2}$$

(c) Normalized pattern $F(\theta) = S(\theta)/S_{\max}$ (5 pts):

$$F(\theta) = \frac{1}{0.0858} \left[\frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$$

Bell-shaped curve peaking at $\theta = \pi/2$ with $F = 1$, dropping to 0 at $\theta = 0$ and π . Symmetric about $\theta = \pi/2$. Similar shape to half-wave dipole pattern but slightly narrower since the dipole is shorter.

Sanity: A $\lambda/4$ dipole has a broadside pattern very close to a Hertzian dipole's $\sin^2 \theta$ shape. As $l \rightarrow 0$ the bracket approaches $\sin \theta$ and $S \propto \sin^2 \theta$ (Hertzian). Half-wave $l = \lambda/2$ has slightly narrower beam than the $\lambda/4$ case.

Problem 4 — Effective Area vs. Physical Cross-section

5 pts

Setup: Half-wave dipole at $f = 100$ MHz, wire diameter $d = 2$ cm. Find effective area A_e and compare to physical cross-section A_p .

EFFECTIVE AREA A_e

Definition *antenna's RX cross section (m^2)*

$$A_e = \frac{\lambda^2 D}{4\pi} \text{ (m}^2\text{)}$$

Physical cross section *wire dipole, diameter d*

$$A_p = l d = \frac{\lambda}{2} d \text{ (half-wave)}$$

Inverse: D from A_e

$$D = \frac{4\pi A_e}{\lambda^2} \text{ (unitless)}$$

Hertzian special case $D = 1.5$

$$A_e^{\text{Hertz}} = \frac{3\lambda^2}{8\pi} \approx 0.119\lambda^2 \text{ (m}^2\text{)}$$

Thin wire: $A_e \gg A_p$. Antenna captures field over $\sim \lambda$.

COMMON DIPOLE PARAMETERS

Type	D	D_{dB}	$R_{\text{rad}} \text{ (}\Omega\text{)}$
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2 (l/\lambda)^2$
Half-wave	1.64	2.15	73.2
$\lambda/4$ monopole	1.64	2.15	36.6
$l = \lambda$	2.41	3.82	199

Half-wave: $P_{\text{rad}} = 36.6 I_0^2$ W. $\lambda/4$ monopole (above ground): same patt. as half-wave but $P_{\text{rad}}, R_{\text{rad}}$ halved.

Step 1 — Wavelength.

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^8} = 3 \text{ m}$$

Step 2 — Directivity for half-wave dipole. From COMMON DIPOLE PARAMETERS: $D = 1.64$.

Step 3 — Effective area. From EFFECTIVE AREA A_e :

$$A_e = \frac{\lambda^2 D}{4\pi} = \frac{(3)^2 (1.64)}{4\pi} = \frac{14.76}{4\pi}$$

$$A_e = 1.17 \text{ m}^2$$

Step 4 — Physical cross-section. A half-wave dipole has length $l = \lambda/2$. Treat as a rectangle of length l and width d :

$$A_p = l \cdot d = \frac{\lambda}{2} \cdot d = \frac{3}{2}(0.02) = 0.03 \text{ m}^2$$

Step 5 — Ratio.

$$\frac{A_e}{A_p} = \frac{1.17}{0.03} = 39.2$$

Sanity: The effective area is $39\times$ larger than the physical cross-section. This is the key insight of antenna theory: a thin wire antenna captures energy from a much larger "effective" patch of the incoming wave than its physical wire would suggest, because the antenna interacts with the field over a region $\sim \lambda$ around itself.

Problem 5 — Friis Transmission, TV Broadcast

5 pts

Setup: Half-wave dipole TX, 1 kW at 50 MHz. RX home antenna with 3 dB gain, distance 30 km. Find received power.

FRIIS TRANSMISSION FORMULA

Far-field link budget P_t tx, P_r rx, range R

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \quad (\text{W})$$

 G_t, G_r are linear gains; $\lambda = c/f$; R in m; P_t, P_r in W.Solve for R max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \quad (\text{m})$$

Equivalent form using A_e recv. area form

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

General form (off-axis) when patterns NOT peak-aligned

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

 F_t, F_r normalized patterns at the link directions; $F = 1$ when aligned.Recipe 1. $\lambda = c/f$. 2. Convert dB gains to linear: $G = 10^{G_{dB}/10}$.3. If TX is “half-wave dipole” use $G_t = 1.64$. 4. Plug into Friis.Use linear G , not dB. Convert FIRST.

dB ↔ LINEAR

$$G_{dB} = 10 \log_{10} G_{lin} \quad G_{lin} = 10^{G_{dB}/10}$$

dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
-3	0.5	-10	0.1

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.For E-field / amplitude use 20 log not 10 log (power $\propto E^2$).

Step 1 — Wavelength.

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{5 \times 10^7} = 6 \text{ m}$$

Step 2 — Antenna gains. TX is half-wave dipole, so $G_t = 1.64$ (from COMMON DIPOLE PARAMETERS).

RX gain: 3 dB $\rightarrow$ linear via $G_{lin} = 10^{G_{dB}/10}$:

$$G_r = 10^{3/10} = 2$$

(3 dB always = $\times 2$, this is the cleanest dB conversion to remember.)**Step 3 — Friis formula.** From FRIIS TRANSMISSION FORMULA:

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t$$

Step 4 — Plug in. $R = 30 \text{ km} = 3 \times 10^4 \text{ m}$, $P_t = 10^3 \text{ W}$:

$$P_r = (1.64)(2) \left(\frac{6}{4\pi(3 \times 10^4)} \right)^2 (10^3)$$

The path-loss factor:

$$\frac{\lambda}{4\pi R} = \frac{6}{4\pi \cdot 3 \times 10^4} = \frac{6}{3.77 \times 10^5} = 1.59 \times 10^{-5}$$

$$\left(\frac{\lambda}{4\pi R} \right)^2 = 2.53 \times 10^{-10}$$

$$P_r = (1.64)(2)(2.53 \times 10^{-10})(10^3) = (3.28)(2.53 \times 10^{-7})$$

$$P_r = 8.3 \times 10^{-7} \text{ W} = 0.83 \mu\text{W}$$

Sanity: 1 kW transmitted, 0.83 μW received at 30 km. That’s a 90 dB drop, dominated by the $1/R^2$ free-space path loss. Plenty of signal for a TV receiver (typical sensitivity is $-100 \text{ dBm} = 10^{-13} \text{ W}$), so this link closes comfortably.

Note on the solution PDF: the official key wrote “(1.64)(3)” as a typo, but used $G_r = 2$ in the actual arithmetic (since 3 dB $\rightarrow$ linear = 2). The final answer $8.3 \times 10^{-7} \text{ W}$ matches either way.

Problem 6 — Communication Link with SNR

15 pts (new)

Setup: $G_t = 20$ dB, $G_r = 23$ dB, $R = 20$ km, $P_t = 10$ W, $f = 6$ GHz. All matched. Find: (a) power density at RX antenna, (b) received power, (c) SNR in dB if $T_{\text{sys}} = 1000$ K and $B = 20$ MHz.

FRIIS TRANSMISSION FORMULA

Far-field link budget P_t tx, P_r rx, range R

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \quad (\text{W})$$

 G_t, G_r are linear gains; $\lambda = c/f$; R in m; P_t, P_r in W.Solve for R max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \quad (\text{m})$$

Equivalent form using A_e recv. area form

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

General form (off-axis) when patterns NOT peak-aligned

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

 F_t, F_r normalized patterns at the link directions; $F = 1$ when aligned.Recipe 1. $\lambda = c/f$. 2. Convert dB gains to linear: $G = 10^{G_{\text{dB}}/10}$.3. If TX is "half-wave dipole" use $G_t = 1.64$. 4. Plug into Friis.Use linear G , not dB. Convert FIRST.

NOISE & SNR

Noise power thermal noise into matched receiver

$$P_N = k_B T_{\text{sys}} B \quad (\text{W})$$

 $k_B = 1.38 \times 10^{-23}$ J/K; T_{sys} system noise temp (K); B receiver bandwidth (Hz).

Signal-to-noise ratio

$$\text{SNR} = \frac{P_{\text{rec}}}{P_N} \quad (\text{unitless}) \quad \text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$$

Typical comm. link needs SNR > 10 dB for usable signal.

dB $\leftrightarrow$ LINEAR

$$G_{\text{dB}} = 10 \log_{10} G_{\text{lin}} \quad G_{\text{lin}} = 10^{G_{\text{dB}}/10}$$

	dB	linear	dB	linear
	0	1	10	10
	3	2	13	20
	6	4	20	100
	7	5	30	1000
	-3	0.5	-10	0.1

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.For E-field / amplitude use $20 \log$ not $10 \log$ (power $\propto E^2$).

Setup numbers.

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{6 \times 10^9} = 0.05 \text{ m}$$

$$G_t = 10^{20/10} = 100 \quad G_r = 10^{23/10} = 199.5$$

(a) Power density at RX antenna. TX radiates $G_t P_t$ over a sphere of area $4\pi R^2$:

$$S_r = \frac{G_t P_t}{4\pi R^2} = \frac{(100)(10)}{4\pi(2 \times 10^4)^2} = \frac{1000}{5.027 \times 10^9}$$

$$S_r = 1.989 \times 10^{-7} \text{ W/m}^2 \approx 0.199 \text{ } \mu\text{W/m}^2$$

(b) Received power (Friis).

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t$$

$$\frac{\lambda}{4\pi R} = \frac{0.05}{4\pi(2 \times 10^4)} = 1.989 \times 10^{-7}$$

$$\left(\frac{\lambda}{4\pi R} \right)^2 = 3.957 \times 10^{-14}$$

$$P_r = (100)(199.5)(3.957 \times 10^{-14})(10) = 7.894 \times 10^{-9}$$

$$\boxed{P_r = 7.89 \text{ nW}}$$

Check: equivalent via effective area: $A_{e,r} = G_r \lambda^2 / (4\pi) = 199.5(0.0025) / (4\pi) = 0.0397 \text{ m}^2$, then $P_r = S_r A_{e,r} = (1.989 \times 10^{-7})(0.0397) = 7.89 \times 10^{-9} \text{ W}$ ✓

(c) Signal-to-noise ratio.

Noise floor (from NOISE & SNR):

$$P_N = k_B T_{\text{sys}} B = (1.38 \times 10^{-23})(1000)(2 \times 10^7) = 2.76 \times 10^{-13} \text{ W}$$

$$\text{SNR} = \frac{P_r}{P_N} = \frac{7.89 \times 10^{-9}}{2.76 \times 10^{-13}} = 2.86 \times 10^4$$

$$\boxed{\text{SNR}_{\text{dB}} = 10 \log_{10}(2.86 \times 10^4) = 44.6 \text{ dB}}$$

Sanity: > 40 dB SNR is a very clean link. Plenty for digital comm (usually > 10 dB suffices).

Problem 7 — Circular Aperture Antenna Scaling

15 pts (new)

Setup: Circular aperture, circular beam $\beta = 3^\circ$ at $f = 20$ GHz. Find (a) D in dB, (b) new D and β if A_p doubled, (c) new D and β if frequency doubled (same A_p).

APERTURE ANTENNAS (horn, dish)

Directivity from aperture area *uniform illumination*

$$D \approx \frac{4\pi A_p}{\lambda^2} \text{ (unitless)} \Rightarrow A_e \approx A_p$$

Directivity from beamwidths *rectangular beam*

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}} \text{ (use rad)}$$

Circular beam ($\beta_{xz} = \beta_{yz} = \beta$): $D \approx 4\pi/\beta^2$.Half-power beamwidth *uniform rect. illumination*

$$\beta_{xz} \approx 0.88 \frac{\lambda}{\ell_x} \text{ (rad)} \quad \beta_{yz} \approx 0.88 \frac{\lambda}{\ell_y}$$

Scaling: $D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$. Double area $\Rightarrow D \times 2$, $\beta/\sqrt{2}$. Double f ($\lambda/2$) $\Rightarrow D \times 4$, $\beta/2$.

dB $\leftrightarrow$ LINEAR

$$G_{\text{dB}} = 10 \log_{10} G_{\text{lin}} \quad G_{\text{lin}} = 10^{G_{\text{dB}}/10}$$

dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
-3	0.5	-10	0.1

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use 20 log not 10 log (power $\propto E^2$).

(a) **Directivity from beamwidth.** For circular beam ($\beta_{xz} = \beta_{yz} = \beta$):

$$D \approx \frac{4\pi}{\beta^2} \text{ (use } \beta \text{ in radians)}$$

$$\beta = 3^\circ \cdot \frac{\pi}{180} = 0.05236 \text{ rad}$$

$$D = \frac{4\pi}{(0.05236)^2} = \frac{12.566}{0.002742} = 4584$$

$$D = 10 \log_{10}(4584) = 36.6 \text{ dB}$$

(b) **Area doubled, $A_p \rightarrow 2A_p$.** Use $D \propto A_p$ (from $D \approx 4\pi A_p/\lambda^2$):

$$D_{\text{new}} = 2D_{\text{old}} = 9168 \Rightarrow D_{\text{new}} = 39.6 \text{ dB}$$

Since $\beta \propto \lambda/\sqrt{A_p} \propto 1/\sqrt{A_p}$:

$$\beta_{\text{new}} = \beta/\sqrt{2} = 3^\circ/1.414 = 2.12^\circ$$

(c) **Frequency doubled ($\lambda \rightarrow \lambda/2$), same A_p .** Use $D \propto A_p/\lambda^2$:

$$D_{\text{new}} = 4D_{\text{old}} = 18336 \Rightarrow D_{\text{new}} = 42.6 \text{ dB}$$

Since $\beta \propto \lambda$:

$$\beta_{\text{new}} = \beta/2 = 1.5^\circ$$

Sanity: bigger area or shorter wavelength both sharpen the beam and increase directivity. Doubling area gives +3 dB ($\times 2$); doubling frequency gives +6 dB ($\times 4$) because both D and $1/\beta^2$ scale.

Problem 8 — 5-Element Linear Array, $d = 3\lambda/4$ **(new)**

Setup: Uniform-amplitude broadside linear array, $N = 5$ elements, equal-phase, interelement spacing $d = 3\lambda/4$. Find (a) normalized array factor, (b) HPBW.

LINEAR ANTENNA ARRAY

Setup N identical elements, spacing d along axis

element pos $z_i = id$, $i = 0, \dots, N-1$; $A_i = a_i e^{j\psi_i}$

Pattern multiplication

$$S(R, \theta, \phi) = S_e(R, \theta, \phi) F_a(\theta)$$

Array factor $F_a(\theta) = \left| \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta} \right|^2$

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta} \right|^2$$

Uniform $A_i = 1$, **equal phase** *closed form*

$$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}, \quad \gamma = kd \cos \theta$$

Normalize: F_a/N^2 . *Peak* N^2 *at* $\gamma = 0$ ($\cos \theta = 0$, *broadside*).

HPBW (broadside, uniform) *array length* $L = (N-1)d$
 $\beta \approx 0.88 \frac{\lambda}{L}$ (rad)

(a) Normalized array factor. From LINEAR ANTENNA ARRAY (uniform, equal-phase):

$$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}, \quad \gamma = kd \cos \theta$$

With $kd = (2\pi/\lambda)(3\lambda/4) = 3\pi/2$:

$$\gamma = \frac{3\pi}{2} \cos \theta, \quad N\gamma/2 = \frac{15\pi}{4} \cos \theta, \quad \gamma/2 = \frac{3\pi}{4} \cos \theta$$

$$F_a(\theta) = \frac{\sin^2\left(\frac{15\pi}{4} \cos \theta\right)}{\sin^2\left(\frac{3\pi}{4} \cos \theta\right)}$$

Peak at $\gamma = 0$ ($\cos \theta = 0$, broadside): $F_a^{\max} = N^2 = 25$.

$$F_a^{\text{norm}}(\theta) = \frac{1}{25} \cdot \frac{\sin^2((15\pi/4) \cos \theta)}{\sin^2((3\pi/4) \cos \theta)}$$

(b) Half-power beamwidth. For uniform broadside array, the standard result is:

$$\beta \approx 0.886 \frac{\lambda}{Nd} \text{ (rad)}$$

$$\beta = 0.886 \cdot \frac{\lambda}{5(3\lambda/4)} = 0.886 \cdot \frac{4}{15} = 0.236 \text{ rad}$$

$$\beta \approx 13.5^\circ$$

Check by solving: $F_a^{\text{norm}} = 0.5$ at $\gamma/2 \approx 0.28$ rad, so $\cos \theta = 0.28/(3\pi/4) = 0.119$, $\theta = 83.2^\circ$, offset from broadside (90°) by 6.8° , full HPBW $\approx 13.6^\circ$ ✓.

Sanity: 5 closely-spaced elements give a moderately narrow beam ($\sim 14^\circ$) compared to a single dipole's $\sim 78^\circ$. Adding more elements or increasing spacing further narrows the main lobe (with grating-lobe risk if $d > \lambda$).